

- 1) Science is qualitatively distinct from, superior to and ideal for all other areas of human endeavour (**scientism**).
- 2) The distinction, superiority and idealhood that science enjoys is traceable to its possession of a method (**Methodologism**).
- 3) There is only one method common to all science irrespective of their subject matter (Methodological Monism).
- 4) That method which is common to all sciences, natural and human, is the method of Hypothetico-Deduction (**Hypothetico-Deductivism**).
- 5) The hallmark of science (i.e., the distinguishing mark of science) consists in the fact that its statements are systematically falsifiable (**falsifiability**).
- 6) Scientific observations are not and cannot be shown to be pure; that is, they are theory-dependent.
- 7) Theories are not winnowed from observations or facts; they are pure inventions of human mind i.e., only conjectures and not generalizations based on 'pure observations'.
- 8) The relation between observation and theory is one of interdependence.
- 9) To a given set of observation-statements there might correspond more than one theory.
- 10) Our factual judgements may have value commitments and our value judgements may have cognitive content (hence fact-value dichotomy is unacceptable); science is not value neutral but the value commitments can be critically discussed and therefore they are not subjective.
- 11) All scientific explanations must have deductive-nomological pattern and thus the thesis of Deductive-Nomologism is acceptable.
- 12) The aim of science is to provide an account of observable world in terms of unobservable entities and to provide accounts of those unobservable entities in terms of further unobservable entities. Unobservable entities are, therefore, real and our theories are putative descriptions of such real entities ('Realism').
- 13) Unlike other areas of human activity, there is progress in science which consists in going from one theory to a better theory. Here, 'better' means 'more true'. 'More to true' means 'the world of unobservables'. In short, science is progressive in the sense our successive theories in any domain of science exhibit greater and greater verisimilitude or truth-nearness i.e. the match between our theories and reality. Unlike positivists, Popper rejects the idea that progress of science is characterized by cumulative growth of theories. According to him, a new theory is entirely new and not an old theory plus an epsilon as Positivists thought. Thus, in Popper's scheme, the growth of science is essentially discontinuous. Of course, Popper makes some room for continuity also when he says that old theory (at least true part of it) is a limiting case of the new theory.
- 14) Science is not objective in the sense scientific theories are based on pure observations as positivists thought because there are no pure observations. Science is objective in the sense its theories are inter-subjectively testable.

- 15) Lastly, science is not rational in the sense the principle of Induction can be rationally justified as Positivists thought. The principle of Induction cannot be rationally justified; nor is it used by science. Science is rational in the sense it embodies critical thinking. Apart from insisting that our theories be falsifiable, science has institutional mechanisms for practicing and promoting critical thinking. What is rationality other than critical thinking? Positivists and Popper differ from each. The theses (1), (2), (3) and (11) are common to both Positivists and Popperians. Popper rejects most of other theses of Positivists, especially their central thesis which concerns the idea of pure observation. Finally, he agrees with the Positivists that science is uniquely progressive, objective and rational; but his notions of progressiveness, objectivity and rationality of science are entirely different from those of Positivists.

2.5 CRITICISM AGAINST KARL POPPER'S PHILOSOPHY OF SCIENCE

A serious lacuna in Popper's position concerns his idea of scientific progress.

First of all, according to Popper, the growth of science is essentially discontinuous in the sense that a new theory which displaces an old theory is not the old theory plus an epsilon because it is entirely new. Yet, he seeks to make room for continuity in the growth of science by insisting that the old theory is a limiting case of the new theory. In this connection he cites an example of Newtonian mechanics and Relativistic mechanics. The former is the limiting case of the latter in the sense that in a certain domain both give the same results. Thus the former is contained in the latter. Hence there is some continuity in the growth of science. But Popper overlooks the fact that such examples of an old theory being a limiting case of the new one are rare. For example, it is absurd to say that Phlogiston theory is a limiting case of oxygen theory or that Ptolemy's theory is limiting case of Copernican theory.

Secondly, Popper says that successive theories in any domain exhibit increasing verisimilitude i.e., truth nearness. That is, reality constituted by unobservable entities is more like what a new theory says than what its immediate predecessor says. This means that following Popper we have to say that the ultimate constituents of matter are more like fields as the present physical theory says than like particles (atoms) as claimed by Newtonian theory. This is unintelligible. What does it mean to say that the ultimate constituents of matter are more like fields than particles called atoms? Either they are like fields or like particles.

Thirdly, when Popper says a new theory is better than the old one (in the sense it is more true), he assumes that the two theories can be compared. This means that they have something common which makes them comparable. But this has been ably questioned by Thomas Kuhn who sought to show that when one fundamental theory replaces another, the two theories are so radically different as to make any talk of comparison between them highly questionable. We shall now turn to his views.

Check Your Progress 2

- 1) On what basis, according to Popper, a line can be demarked between science and rest of knowledge:

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- 2) Identify the main differences between Inductive Model and Hypothetico-Deduction Model.

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- 3) What are the Central components of Hypothetico-Deductive Model?

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- 4) In what sense, science is rationale?

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2.6 THOMAS KUHN'S PHILOSOPHY OF SCIENCE

Thomas Kuhn's work **The Structure of Scientific Revolutions** is a milestone in the history of the 20th century of philosophy of science. A brief exposition of his basic ideas are given below.

According to Kuhn, in the life of every major science there are two stages (1) **Pre-paradigmatic stage**, and (2) **paradigmatic stage**. In the pre-paradigmatic stage one finds more than one mode of practicing that science. That is, there was a time in the history of Astronomy when different schools of Astronomy practised Astronomy differently. So is the case with Physics, Chemistry and Biology. In that stage their situation was similar to that which obtains today in areas like art, philosophy and even medicine wherein divergent modes of practicing these disciplines co-exist. Today, we speak of schools of Art (e.g., painting), schools of Philosophy and systems/schools of medicine. But today we do not speak of Schools of Astronomy or Physics or Chemistry or Biology.

This is, according to Kuhn, in areas like art, philosophy and medicine that did not, and cannot make a transition from pre-paradigmatic stage to paradigmatic stage, which marks the disappearance of plurality, that is, disappearance of schools. In other words, the transition means replacement of plurality by monolith i.e., a uniform mode of practice.

Such a transition is made possible, Kuhn claims, by acquisition of a **paradigm**. When a science makes such a transition, we may say, it has become 'mature' or 'science' in the proper sense of the term. Astronomy was the first to make such a transition followed by Physics, Chemistry and Biology in that order. Social Sciences are still, according to him, in the pre-paradigmatic stage, though Economics is showing signs of such a transition. This is evident from the fact that in Social Sciences there is no consensus on fundamentals as we can see prevalence of distinct schools in every Social Science.

So, the transition to maturity is effected by acquiring a paradigm by a science. The question is "What is a paradigm?"

We all know that Ptolemy's *Almagest* Newton's *Principia* and Darwin's *Origin of the Species* are path-breaking works in the areas of Astronomy, Physics and Biology respectively. According to Kuhn, these works provided paradigms for these disciplines. They did so by specifying the exact manner in which these disciplines ought to proceed. They laid the ground rules regarding what problems these disciplines must tackle and how to tackle them. Hence, paradigms are Universally recognized achievements that for a time provide model problems and solutions to community of practitioners. Hence, in the **first** place, a paradigm specifies what that ultimate constituents of that sphere of reality, which a particular science is inquiring into, are.

Secondly, it identifies the model problems. **Thirdly**, it specifies the possible range of solutions. **Fourthly**, it provides the necessary strategies and techniques for solving the problems. **Lastly**, it provides examples which show how to solve certain problems. In other words, a paradigm is a disciplinary matrix of a professional group. Once a science possesses a paradigm, it develops what Kuhn calls, a 'normal science tradition'. Normal science is the day-to-day research activity purporting to force of nature into conceptual boxes provided by the paradigm. The practitioners of normal science, that is, a scientist who engages in day-to-day research, internalizes the paradigm by professional education. This explains the prevalence of textbook culture in science education.

Of course, scientific practice is not exhausted in terms of day-to-day research or 'normal science'. When a paradigm fails to promote fruitful, interesting and smooth normal science, it is considered to be in a crisis. The deepening of the crisis leads to the replacement of the existing paradigm by a new one. This process of replacement is called '**scientific revolution**'. Therefore, scientific revolutions are "the tradition-shattering complements to the tradition bound activity of normal science." Thus, once a science enters the paradigmatic stage, it is characterized by (1) normal science, and (2) revolutions. In sheer temporal terms, normal science occupies much larger span than revolutionary science. That is to say, science is revolutionary once a while and mostly it is non-revolutionary or normal. Also the scientific activity engaged in by most of the practitioners can be characterized aptly in terms of normal science. On account of this temporal and numerical

magnitude we can say that much of the scientific activity as we ordinarily encounter is normal though this normal course is occasionally interrupted by revolutions which change the form, content and direction of the process of the scientific activity, which is basically normal by which we mean a non-revolutionary committed and tradition bound activity. Normal science demands a through going convergent thinking and hence is preceded by an education that involves 'a dogmatic initiation in a pre-established tradition that the student is not equipped to evaluate'. "Normal science is an activity that purports not to question the existing paradigm but to (1) "Increase the precision of the existing theory by attempting to adjust the existing theory or existing observation in order to bring the two into closer and closer agreement", and (2) "to extend the existing theory to areas that it is expected to cover but in which it has never before been tried". In other words, normal science consists of solving puzzles that are encountered in forcing nature into the conceptual boxes supplied by the reigning paradigm.

It is in this way Kuhn attempts to account for the smooth, defined and directional character of day-to-day scientific research in terms of the features of what he calls "Normal Science". Normal Science has no room for any radical thinking. It is limited to the enterprise of solving certain puzzles in accordance with the rules specified by the paradigm. These rules are never questioned but only accepted and followed. The aim of scientific education is to ensure that the paradigm is internalized by a student. In other words, the professional training in science consists in accepting the paradigm as given and equipping oneself to promote the cause of paradigm by giving a greater precision and further elaboration. The day-to-day scientific research does not aim at anything fundamentally new but only at the application of what has already been given, namely the theoretical ideas and the practical guidelines for solving certain puzzles. It is in this sense that **normal science is a highly tradition-bound activity**.

As pointed out earlier, normal science purports to force nature into the conceptual boxes provided by the reigning paradigm by solving puzzles in accordance with the guidelines provided by the paradigm whose validity is accepted without question. During this process of puzzle solving, certain hurdles may be encountered. We then speak of "anomalies". That is, an anomaly arises when a puzzle remains puzzle defying every attempt to resolve it within the framework of the paradigm. But, appearance of one or two anomalies is not sufficient to overthrow a paradigm. The ushering in of the era of a new paradigm has to be preceded by the appearance of not one or two minor anomalies, but many major ones. In order to declare a paradigm to be crisis-ridden, what is needed is an accumulation of major anomalies. But there is no clear cut and objective criterion to decide which anomalies are major and how many anomalies must accumulate to declare a paradigm to be crisis-ridden. In other words, there is no criterion which decides whether a perceived anomaly is only a puzzle or the symptom of a deep crisis. The issue will be decided by the community of the practitioners of the discipline through the judgment of its peers. Once the scientific community declares the existing paradigm to be crisis-ridden, the search for the alternative begins. Of course the crisis-ridden paradigm will not be given up unless a new theory is accepted in its place. It is only during this transitional period of search for the new paradigm that the scientific debates become radical.

During the process of search for an alternative, the scientific community has to make a choice between competing theories. In this choice, the evaluation procedures of normal science are of no help, “for these depend in part upon a particular paradigm and that paradigm is at issue.” The issue concerning the paradigm choice can not be settled by logic and experiment alone. What ultimately matters is the consensus of the relevant scientific community. Kuhn points out, “that question of value can be answered only in terms of criteria that lie outside of normal science altogether, and it is that recourse to external criteria that most obviously makes paradigm debates revolutionary.” Thus a paradigm choice cannot be explicated in the neutral language of mathematical equations and experimental procedures, but in terms of specific perceptions which a scientific community as a social entity entertains about what it considers to be the basic values of its professional enterprise. In other words, the ultimate explanation of a theory choice is not methodological but sociological. Hence in Kuhn’s scheme, the idea of scientific community as a social entity is axiomatic. That is to say, according to him, “If the term “paradigm” is to be successfully explicated, scientific communities must first be recognized as having an independent existence”. This means that one must explain scientific practice in terms of paradigms and paradigmatic changes and the latter are to be explicated in terms of a particular scientific community which shares the paradigms and brings about paradigmatic changes. Thus, the concept of scientific community is basic to the concept of paradigm. The concept of **scientific community** can be explicated in sociological terms. Hence, the ultimate terms of explication of scientific activity are sociological.

What is the relation between the old paradigm, which is overthrown and the new paradigm which succeeds it? Kuhn’s answer to this question is extremely radical. According to him, in no obvious sense one can say that the new paradigm is better or truer than the old one. Kuhn maintains that the two successful paradigms cut the world differently. They speak different languages. In fact when a paradigm changes, to put it metaphorically, the world changes. With his characteristic lucidity he says, “the transition from a paradigm in crisis to new one from which a new tradition of normal science can emerge is far from accumulative process, one that is achieved by an articulation or extension of the old paradigm. Rather, it is a reconstruction of the field from new fundamentals, a reconstruction that changes some of the field’s most elementary theoretical generalizations as well as many of its,... methods and applications.” This apart Kuhn contends that the two paradigms talk different languages. Even if the same terms are used in two paradigms, the terms have different meanings. What can be said in the language of one paradigm can not be translated into the other language. Based on these reasons, Kuhn claims that the relation between two successive paradigms is incommensurable. No wonder Kuhn compares paradigm shift to gestalt shift. With this, the idea of scientific progress as a continuous process and the idea of truth as the absolute standard stand totally repudiated. Kuhn advances what might appear to be an undiluted relativism according to which truth is intro-paradigmatic and not inter-paradigmatic. That is to say what is true is relative to a paradigm and there is no truth lying outside all paradigms.

2.7 POPPER VERSUS KUHN

Some of the radical implications of Kuhn's position can be brought about by juxtaposing his views with those of Popper.

Firstly, the hallmark of science according to Popper is **critical thinking**. In fact science exemplifies critical thinking at its best. Since critical thinking considers nothing to be settled and lying beyond all doubt, fundamental disagreements and divergent thinking must and in fact do characterize science. As we have seen, according to Kuhn, what constitutes the essence of scientific practice is normal science and we have also seen why normal science is a highly tradition-bound activity, an activity made possible by a consensus among the practitioners who share a paradigm. Thus if Popper sees the essence of science in divergent thinking and fundamental disagreements, Kuhn sees the essence of science in convergent thinking and consensus. In other words, the hallmark of science according to Kuhn is tradition-bound thinking. In fact, according to Kuhn, what distinguishes science from other areas of creative thinking is that whereas in science one finds institutional mechanisms of enforcing consensus, the other areas suffer from perpetual disagreements even on fundamentals.

Secondly, if **Popper** considers the individual to be the focus of scientific activity, Kuhn bestows that status upon the scientific community. Both positivists and Popper looked upon science as the sum total of the work of individual scientists working in accordance with a method though the Positivists and Popper fundamentally differed on the characterization of that method. As opposed to this individualistic account of scientific enterprise, Kuhn propounds a collectivistic view of scientific activity. In Kuhn's scheme, it is the scientific community, which constitutes the pillar of stability and locomotive of change. This is borne out by the fact that according to Kuhn the scientific community has institutional mechanism, like peer review, by which it can settle all the issues such as whether an anomaly is a symptom of crisis, how many anomalies suffice to warrant the search for an alternative paradigm, what factors are to be considered in choosing a new theory for the status of the paradigm etc.

Thirdly, Popper and Kuhn differ fundamentally in their attitude towards the transition from one theory to another theory in science. According to Popper, we can explain every case of change of theoretical framework in terms of certain norms which science always adopts and follows meticulously. In fact, scientific rationality consists in following these norms. But Kuhn contends that an adequate explanation of change of theoretical framework must be in terms of the value judgments made by a community while making the choice. According to Kuhn, recourse to the so-called methodological norms explains nothing. From the point of view of Popper, Kuhn is an irrationalist because he sets aside methodological norms and seeks to explain theory change exclusively in terms of non-rational or sociological factors like value commitments of a professional group. Whatever be the merit of Popper's attack on Kuhn as an irrationalist, we can say that Kuhn's construct of scientific practice is sociological. That is to say, according to him, scientific activity can not be understood by trying to find out the absolute standards which have guided the scientific activity in all ages. It can only be understood in terms of the specific judgments which a community makes at a particular juncture regarding what it considers to be its value commitments as a professional group.

The above juxtaposition of Popper and Kuhn brings out the radical implications of Kuhn's views regarding the nature of scientific practice. However, in one respect Kuhn is very close to Popper. Both, like positivists, content that there is something unique to science though they differ in their explanation of what that uniqueness consists of. Positivists maintain that the hallmark of science is the systematic verifiability of its claims. According to Popper, the uniqueness of science consists in the systematic falsifiability of theories. According to Kuhn, it is consensus which marks out science from other areas of human endeavour. That is to say, Kuhn, like Positivists and Popper, does not question whether science is really unique. That is to say, instead of raising critical questions about the status that science has acquired in the contemporary culture, Kuhn only seeks to provide an alternative account of how it has acquired that status. In that sense Kuhn's position is quite conservative.

In this unit, we had a brief look at the 20th century thinking on the nature of science. It is very difficult to decide which view is the correct one, though Positivist view has been shown to be highly inadequate. The question is "How should we practise social sciences so as to make them scientific"?

Some social scientists take the Positivist recommendation: "collect data, extract a generalization, verify the generalization and formulate a law". Those social scientists who are inspired by the Popperian view, take seriously the Popperian advice "Formulate a problem, provide a tentative solution, try to falsify it, if the solution survives treat it as a corroborated theory but not as a confirmed one". Still others go by Kuhn's view of science and think that the task of social sciences today is to arrive at paradigms in different social scientific disciplines. This will enable social sciences to overcome ideological commitments which generate differences even at a fundamental level. According to them, the consensus so generated will bring social sciences close to natural sciences.

Check Your Progress 3

- 1) Which two stages come in the life of every major science?

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- 2) What do you understand by the term 'Paradigm'?

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3) What is scientific revolution?

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4) How did Popper differ with Kuhn about essence of science?

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5) Match the following:

A	B
1) Individual as the locus of scientific study	i) Positivists
2) Explanation of theory in terms of value judgment	ii) Popper
3) Theory changes in terms of certain norms	iii) Kuhn
4) Verification as the hall mark of science	iv) Popper

2.8 LET US SUM UP

By virtue of shaping our mode of living and our ways of thinking about the world, science has acquired a central place in intellectual world. Three disciplines namely history of science, sociology of science and philosophy of science have made science itself their object of inquiry. Philosophy of science deals with broadly two questions: what is the aim of the science and what are the methods of science? Different scholars have answered these questions differently.

Regarding aims and methodology two views have been dominant. ‘inductivism’ by Francis Bacon and ‘Hypothesisism’ by Rene Decartes. Positivism has been a movement in the philosophy of the first half of the 20th Century. Inductivism, systematical verifiable statements, pure scientific observations, unilateral relationship between observation and theory, value neutral theory, precise prediction of facts as aim of science, progressive/objective/rational arguments as essentials of science are some of the important central tenets of positivism. The concept of pure observation in positivist philosophy has been attacked on several grounds.

Karl Popper and Thomas Kuhn are the two most important philosophers of mid 20th century. They offered alternatives to positivism and shaped the contemporary debate on theory of scientific knowledge. Popper rejected the central thesis of positivism i.e., idea of ‘pure observation’. He propounded the Hypothetico-Deductive model as an adequate and superior model of scientific method. According to him, the scientific statements are systematically falsifiable. The scientific observations are theory-dependent. Theories are pure inventions of human mind. The theory and observation are interdependent. Science is not value

neutral and therefore values are not subjective. Science is progressive. Science is objective in the sense that its theories are inter-subjectivity testable. Science, according to Popper, is rational in the sense that it embodies critical thinking. Owing to commonality of views of Popper with those of positivists regarding scientism, methodologism, methodological monism and Deductive Nomologism, Popper is sometimes viewed as semi-positivist.

Karl Popper's ideas of scientific progress, verisimilitude and comparison of old theory with new theory have been criticized.

Thomas Kuhn sees the life of major sciences in two stages – Pre-paradigmatic stage and paradigmatic stage. In the pre-paradigmatic stage, more than one mode are practised. In the process of transition a discipline acquires the stage of paradigm. When a paradigm fails to promote fruitful, interesting and smooth normal science, It is considered in crisis. Deepening of crisis leads to replacement of the existing paradigm, which is termed as scientific revolution.

Thus, on the question how we should practise social sciences so as to make them 'scientific, three views have so far been propounded. Positivist recommendation is to 'collect data, extract generalization, verify the generalization and formulate a law'. Popper's view is "formulate a problem, provide a tentative solution, try to falsify it, if solution survives, treat it as a corroborated theory but not as a confirmed one'. Which one out these three is correct – is difficult to decide? Consensus on the three models will bring social science close to natural sciences.

2.9 KEY WORDS

Inductivism

- : A scientific method of knowledge focusing that scientific knowledge is generated by arriving at generalizations on the basis of pure observations using the principle of induction.

Hypothesisism

- : Hypothesisism is a method of knowledge which envisages that knowledge is created or generated by putting forth the hypothesis having test implications. Hypothesis is a statement describing the unobservable entities which lie behind observations.

Realists

- : Those who support Hypothesis are realist in connection with the status of theoretical/unobservable entities.

Anti-Realists

Inductivists are also called as Anti-Realists.

Paradigm

- : Paradigm refers to the scientific practice comprising of a particular a world view, concept, assumptions, theories, models used by the researcher in carrying out the research studies. The concept was introduced by Thomas Kuhn.

Deductive

- : Derivation of a conclusion by reasoning or by inference in which the conclusion follows necessarily from the premises.

- Methodological Dualism** : Referring to positivist belief in separation of the knower from the known or of the subject from object.
- Methodological Monism** : In contrast to the compartmentalisation of dualism, monism view the world as a “seamless web”. In terms of Gouldner’s argument, the separation between the knower and the known must be overcome.
- Pseudo Science** : Pseudo science refers to any body of knowledge or practice, which purports to be scientific or supported by science but which is judged to fall outside the domain of science.

2.10 SOME USEFUL BOOKS

Mark Blang (1992): *The Methodology of Economics*, Cambridge University Press, London.

Uma Devi. S. (1994): *Economic Theory and Methods of Reasoning*, Har-Anand Publication, New Delhi.

Williams, M., & May, T. (1996): *Introduction to the Philosophy of Social Research*, London: UCL Press.

Kuhn, Thomas, (1970); *The Structure of Scientific Revolutions*. The University of Chicago Press: Chicago.

Popper, Karl (1972); *Conjectures and Refutations: The Growth of Scientific Knowledge*, Rutledge and Kegan Paul: London.

2.11 ANSWERS OR HINTS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

- 1) The aim of science is two fold: Definition and Demonstrations.
- 2) See Sub-section 2.2.1
- 3) See Sub-section 2.2.1
- 4) A B
 i) iv)
 ii) i)
 iii) ii)
 iv) iii)
- 5) See Section 2.3

Check Your Progress 2

- 1) On the basis of falsifiability of scientific theories.
- 2) Inductivist model's basic proposition is that observations are pure i.e. theory – independent and theories are derived from pure observations. In Hypothetico-deductive model, theories are perceived as our mental (own) constructions and are not the functions of pure observations.
- 3) Hypothetico-Deductive model's central components are: (i) theories are our own constructions reflecting unobservable entities, (ii) science is progressive in terms of increasing very similitude, (iii) science is rational in terms of critical rationalism and objective in terms of inter-subjectivity.
- 4) Science embodies critical thinking.

Check Your Progress 3

- 1) (i) Pre-paradigmatic stage, (ii) Paradigmatic stage
- 2) The constituents of reality, model problems, range of solutions, necessary strategies and techniques for solving the problems together constitute the paradigm.
- 3) The process of replacement of existing paradigm is known as 'scientific revolution'.
- 4) According to Popper, the essence of science lies in divergent thinking and fundamental disagreements whereas Kuhn sees essence of science in convergent thinking and consensus.
- 5) (1) ii, (2) iii, (3) iv, (4) – i.